

● Hackathon Production Specification • Sept 2026

UniPods Group Memory Platform

An in-group WhatsApp knowledge indexing engine designed for team collaboration, Microsoft Teams meeting transcript ingestion, and admin-prioritized contextual search.

100%

In-Group Native Interaction

MS Teams

OneDrive Recording Sync

Sub-Sec

Vector Knowledge Retrieval

TARGET DEPLOYMENT

Hackathon Judging Phase (Sept 18 - Sept 24)

CORE INTEGRATION

WhatsApp Web API + MS Graph + Supabase

Operational Context & Problem Statement

Fast-moving hackathon teams suffer from fragmented information across chat streams, missed Microsoft Teams meeting recordings, and lost admin directives.

The Problem

Hackathon team members regularly miss live Microsoft Teams calls or forget crucial decisions announced in busy WhatsApp groups. Critical admin announcements get buried under hundreds of chat messages, causing misaligned tasks and missed deadlines.

The Solution

UniPods operates entirely inside your team's WhatsApp group. It continuously indexes group conversations, automatically fetches and transcribes Microsoft Teams recordings from OneDrive, and answers natural language queries with verified admin citations.

Core Design Principles

Key architectural rules governing the UniPods platform built for team efficiency.

- 1 Native Group Experience (No DMs / No External Portals)**
 The bot resides directly inside the team WhatsApp group. Members @mention the bot to search memory, request meeting recaps, or verify admin decisions directly in context.
- 2 Automated Microsoft Teams Call Synchronization**
 Teams meetings recorded by members are automatically synced from Microsoft OneDrive/Teams cloud storage via Graph API, transcribed with Whisper, and added to group memory.
- 3 Admin Authority & Quoting Mechanism**
 The system tags and prioritizes messages sent by group admins. When responding to questions regarding project deadlines or scope, the system explicitly quotes the exact admin who made the decision.
- 4 Privacy-First & Secure Storage**
 Only group messages and authorized Microsoft Teams recordings are indexed into isolated Supabase vector stores. Data is never shared across external environments.

In-Group Interaction Simulation

Real-world scenario: A team member queries missed Microsoft Teams call details and admin directives directly inside WhatsApp.


Hackathon Pod Alpha (UniPods Active)

Elissa, Sarah (Admin), David, UniPods Assistant

Group Chat


Sarah Jenkins

ADMIN

Team reminder: The final submission deadline is strictly Thursday at 20:00 UTC. Make sure your API routes are tested!

14:15

David Chen

I just uploaded the recording from our Teams design sync:
https://m365.sharepoint.com/rec_0917.mp4

16:30

Elissa

@UniPods What did we decide in today's Teams meeting regarding the database schema, and when is our final deadline?

18:42

UniPods Knowledge Assistant


Admin Announcement (Sarah Jenkins):
"The final submission deadline is strictly Thursday at 20:00 UTC."

Summary of MS Teams Call (16:30 Sync):

- **Schema Decision:** Adopted Supabase pgvector with 1536-dim embeddings for message history.
- **Action Item:** David will implement vector indexing for Teams transcriptions by tomorrow morning.

18:42

Key Features Demonstrated

- **Admin Badging:** Clear visual citation of group admin directives.
- **Teams Sync:** Automatic parsing of audio/video recordings.
- **Contextual Synthesizer:** Combines chat history + call transcripts in a single concise reply.

Zero Friction Setup

No app installations required for team members. Simply add the UniPods bot number to the WhatsApp group, assign group admin privileges, and start collaborating.

End-to-End System Architecture

Data ingestion pipeline from WhatsApp and Microsoft Teams into high-performance vector storage and context engine.



Core Technical Stack

Production-grade stack engineered for high speed, reliability, and precision during hackathon judging.

COMPONENT	TECHNOLOGY	ROLE & CAPABILITIES
WhatsApp Gateway	Baileys Protocol API	Read-only group message listener, admin role detection, and metadata extraction.
WhatsApp Delivery	WhatsApp Cloud API	Reliable, official message delivery back to WhatsApp group chat threads.
Meeting Ingestion	Microsoft Graph API	OAuth integration with OneDrive/Teams to fetch `.mp4` and `.m4a` meeting recordings.
Audio Processing	OpenAI Whisper Engine	High-accuracy automated transcription of Teams meeting audio into indexed text chunks.
Vector Database	Supabase (pgvector)	Stores dense embeddings for similarity search across chat logs and call transcripts.
Management UI	Next.js 14 / Tailwind	Team admin dashboard for connected groups, sync status, and storage monitoring.

Hackathon Sprint Schedule (Sept 18 - Sept 24, 2026)

7-day execution plan aligned with hackathon judging milestones.

PHASE 1 • SEPT 18 - SEPT 19

Core Gateway & Admin Sync

- Deploy Baileys listener service.
- Implement admin status detection.
- Setup Supabase vector schema with metadata tags.

PHASE 2 • SEPT 20 - SEPT 21

MS Teams & Speech Ingestion

- Integrate MS Graph API for OneDrive recording fetches.
- Pipeline Whisper speech-to-text processing.
- Enable RAG query engine with admin quote priority.

PHASE 3 • SEPT 22 - SEPT 24

Testing, Dashboard & Judging

- Finalize Next.js admin dashboard.
- End-to-end testing in real group chats.
- Final hackathon demonstration & submission.

Hackathon Success Criteria & Deliverables

1. **100% In-Group Utility:** Zero reliance on third-party web clients during active messaging.
2. **Verifiable Accuracy:** Every answer citing admin announcements includes exact timestamps and quotes.
3. **Seamless Meeting Recaps:** Teams call recordings processed and searchable within minutes of uploading.